

GRIP-5 Circuit Documentation

Summary

This circuit is designed to control five Servo SG90 motors using an Arduino UNO microcontroller. The servos are powered through a step-down buck converter, which is connected to a 9V battery. The Arduino UNO is responsible for sending control signals to each servo motor, allowing for precise movement control. The circuit is suitable for applications requiring multiple servo operations, such as robotics or automated systems.

Component List

1. **Arduino UNO**
 - **Description:** A microcontroller board based on the ATmega328P.
 - **Pins:** UNUSED, IOREF, Reset, 3.3V, 5V, GND, Vin, A0-A5, SCL, SDA, AREF, D0-D13.
2. **Servo SG90 (5 units)**
 - **Description:** A small and lightweight servo motor with a 180-degree range of motion.
 - **Pins:** Signal, Vin, Gnd.
3. **Step Down Buck Converter**
 - **Description:** A device that steps down voltage from a higher level to a lower level.
 - **Pins:** UOUT+, UOUT-, UIN+, UIN-.
4. **9V Battery**
 - **Description:** A standard 9V battery used to power the circuit.
 - **Pins:** Positive, Negative.

Wiring Details

Arduino UNO

- **GND:** Connected to the GND pins of all Servo SG90 motors and the UOUT- pin of the Step Down Buck Converter.
- **D3:** Connected to the Signal pin of Servo SG90 (1).
- **D6:** Connected to the Signal pin of Servo SG90 (2).
- **D9:** Connected to the Signal pin of Servo SG90 (3).
- **D11:** Connected to the Signal pin of Servo SG90 (4).
- **D12:** Connected to the Signal pin of Servo SG90 (5).

Servo SG90 (1)

- **Signal:** Connected to D3 of the Arduino UNO.
- **Vin:** Connected to UOUT+ of the Step Down Buck Converter.
- **Gnd:** Connected to GND of the Arduino UNO and UOUT- of the Step Down Buck

Converter.

Servo SG90 (2)

- **Signal:** Connected to D6 of the Arduino UNO.
- **Vin:** Connected to UOUT+ of the Step Down Buck Converter.
- **Gnd:** Connected to GND of the Arduino UNO and UOUT- of the Step Down Buck Converter.

Servo SG90 (3)

- **Signal:** Connected to D9 of the Arduino UNO.
- **Vin:** Connected to UOUT+ of the Step Down Buck Converter.
- **Gnd:** Connected to GND of the Arduino UNO and UOUT- of the Step Down Buck Converter.

Servo SG90 (4)

- **Signal:** Connected to D11 of the Arduino UNO.
- **Vin:** Connected to UOUT+ of the Step Down Buck Converter.
- **Gnd:** Connected to GND of the Arduino UNO and UOUT- of the Step Down Buck Converter.

Servo SG90 (5)

- **Signal:** Connected to D12 of the Arduino UNO.
- **Vin:** Connected to UOUT+ of the Step Down Buck Converter.
- **Gnd:** Connected to GND of the Arduino UNO and UOUT- of the Step Down Buck Converter.

Step Down Buck Converter

- **UOUT+:** Connected to the Vin pins of all Servo SG90 motors.
- **UOUT-:** Connected to the GND pins of all Servo SG90 motors and the GND of the Arduino UNO.
- **UIN+:** Connected to the Positive pin of the 9V Battery.
- **UIN-:** Connected to the Negative pin of the 9V Battery.

9V Battery

- **Positive:** Connected to UIN+ of the Step Down Buck Converter.
- **Negative:** Connected to UIN- of the Step Down Buck Converter.